

Nmap Cheat Sheet

Basic Scanning Techniques

Scan a Single Target

```
nmap [target]
```

Scan Multiple Targets

```
nmap [target1, target2, etc]
```

Scan a List of Targets

```
nmap -iL [list.txt]
```

Scan a Range of Hosts

```
nmap [range of ip addresses]
```

Scan an Entire Subnet

```
nmap [ip address/cdir]
```

Scan Random Hosts

```
nmap -iR [number]
```

Excluding Targets from a Scan

```
nmap [targets] --exclude [targets]
```

Excluding Targets Using a List

```
nmap [targets] --excludefile [list.txt]
```

Perform an Aggressive Scan

```
nmap -A [target]
```

Scan an IPv6 Target

```
nmap -6 [target]
```

Discovery Options

Perform a Ping Only Scan

```
nmap -sP [target]
```

Don't Ping

```
nmap -PN [target]
```

TCP SYN Ping

```
nmap -PS [target]
```

TCP ACK Ping

```
nmap -PA [target]
```

UDP Ping

```
nmap -PU [target]
```

SCTP INIT Ping

```
nmap -PY [target]
```

ICMP Echo Ping

```
nmap -PE [target]
```

ICMP Timestamp Ping

```
nmap -PP [target]
```

ICMP Address Mask Ping

```
nmap -PM [target]
```

IP Protocol Ping

```
nmap -PO [target]
```

ARP Ping

```
nmap -PR [target]
```

Traceroute

```
nmap --traceroute [target]
```

Force Reverse DNS Resolution

```
nmap -R [target]
```

Disable Reverse DNS Resolution

```
nmap -n [target]
```

Alternative DNS Lookup

```
nmap --system-dns [target]
```

Manually Specify DNS Server(s)

```
nmap --dns-servers [servers] [target]
```

Create a Host List

```
nmap -sL [targets]
```

Advanced Scanning Functions

TCP SYN Scan

```
nmap -sS [target]
```

TCP Connect Scan

```
nmap -sT [target]
```

UDP Scan

```
nmap -sU [target]
```

TCP NULL Scan

```
nmap -sN [target]
```

TCP FIN Scan

```
nmap -sF [target]
```

Xmas Scan

```
nmap -sX [target]
```

TCP ACK Scan

```
nmap -sA [target]
```

Custom TCP Scan

```
nmap --scanflags [flags] [target]
```

IP Protocol Scan

```
nmap -sO [target]
```

Send Raw Ethernet Packets

```
nmap --send-eth [target]
```

Send IP Packets

```
nmap --send-ip [target]
```

Port Scanning Options

Perform a Fast Scan

```
nmap -F [target]
```

Scan Specific Ports

```
nmap -p [port(s)] [target]
```

Scan Ports by Name

```
nmap -p [port name(s)] [target]
```

Scan Ports by Protocol

```
nmap -sU -sT -p U:[ports],T:[ports] [target]
```

Scan All Ports

```
nmap -p "*" [target]
```

Scan Top Ports

```
nmap --top-ports [number] [target]
```

Perform a Sequential Port Scan

```
nmap -r [target]
```

Version Detection

Operating System Detection

```
nmap -O [target]
```

Submit TCP/IP Fingerprints

www.nmap.org/submit/

Attempt to Guess an Unknown OS

```
nmap -O --osscan-guess [target]
```

Service Version Detection

```
nmap -sV [target]
```

Troubleshooting Version Scans

```
nmap -sV --version-trace [target]
```

Perform a RPC Scan

```
nmap -sR [target]
```

Firewall Evasion Techniques

Fragment Packets

```
nmap -f [target]
```

Specify a Specific MTU

```
nmap --mtu [MTU] [target]
```

Use a Decoy

```
nmap -D RND:[number] [target]
```

Idle Zombie Scan

```
nmap -sI [zombie] [target]
```

Manually Specify a Source Port

```
nmap --source-port [port] [target]
```

Append Random Data

```
nmap --data-length [size] [target]
```

Randomize Target Scan Order

```
nmap --randomize-hosts [target]
```

Spoof MAC Address

```
nmap --spoof-mac [MAC|0|vendor] [target]
```

Send Bad Checksums

```
nmap --badsum [target]
```

Output Options

Save Output to a Text File

```
nmap -oN [scan.txt] [target]
```

Save Output to a XML File

```
nmap -oX [scan.xml] [target]
```

Grepable Output

```
nmap -oG [scan.txt] [targets]
```

Output All Supported File Types

```
nmap -oA [path/filename] [target]
```

Periodically Display Statistics

```
nmap --stats-every [time] [target]
```

133t Output

```
nmap -oS [scan.txt] [target]
```

Troubleshooting and Debugging

Getting Help

```
nmap -h
```

Display Nmap Version

```
nmap -V
```

Verbose Output

```
nmap -v [target]
```

Debugging

```
nmap -d [target]
```

Display Port State Reason

```
nmap --reason [target]
```

Only Display Open Ports

```
nmap --open [target]
```

Trace Packets

```
nmap --packet-trace [target]
```

Display Host Networking

```
nmap --iflist
```

Specify a Network Interface

```
nmap -e [interface] [target]
```